Genuine Multipartite Entanglement Analysis — Molecular System

MOLECULE

Methylene (CH2)
Basis set   : cc-pVDZ
Charge   : 0
Multiplicity : 3
Total e-   : 8
n-body order : 3

GROUND STATE ENERGIES

E0 (Ha) : -38.92428881
E1 (Ha) : -38.87208522
Gap (Ha) : 0.05220359
HF (Ha) : -38.92132899

CONVEX-ROOF METHODS

Methods run   : gcg
Restarts   : 8

C_LB : coherence witness bound
C_expo: unitary opt (exp map)
C_stf : Stiefel manifold (Alg. B)
C_UB : eigendecomposition bound

ACTIVE SPACE CAS(6,5)

Frozen core orbs : 1
Frozen e-   : 2
Active e-   : 6
Active orbs   : 5
Spin-orbitals   : 10
Hilbert dim   : 100

SPIN PROPERTIES

<S2>   : 2.000000
S   : 1.0000
<Sz>   : 0.000000
Multiplicity : 3
Spin state   : Triplet (S=1)

GME CRITERIA

C_GME = sqrt(2 * min_A (1 - Tr(rho_A^2)))

Convex roof (mixed):
  C_GME(rho) = min Sum p_i C_GME(|psi_i>)

Bounds ordering:
  C_LB <= C_GME <= C_UB

DOUBLE OCCUPANCY

<D> total : 1.997965
<D> / site : 0.399593

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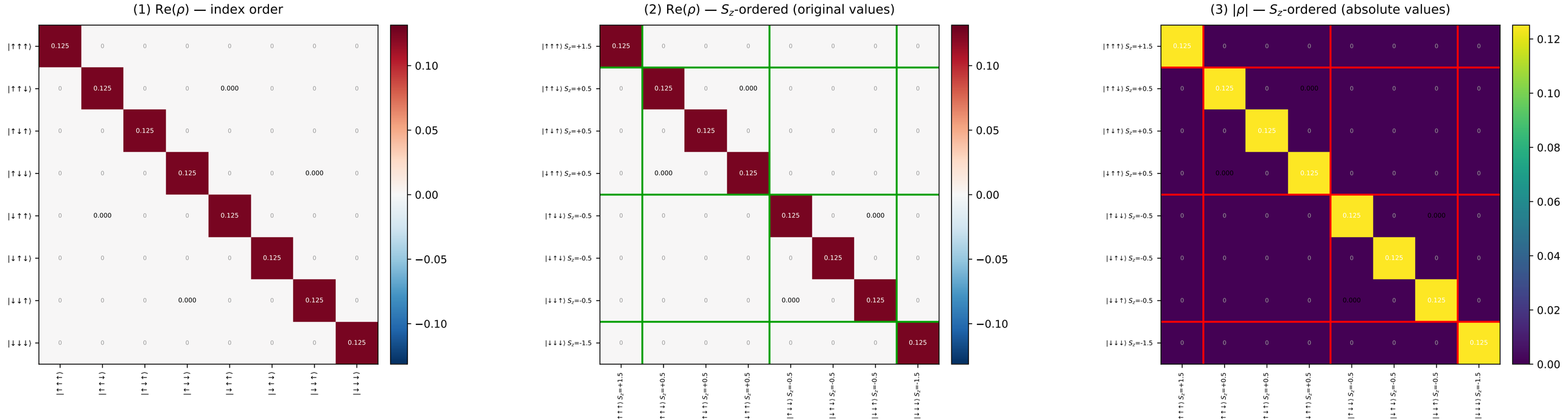
Body order : 3
Clusters   : 10

Date: 2026-07-15 09:32

Summary — 3-Body RDMs (Methylene, CAS(6,5))

Orbitals	Purity	Entropy	C_GME (pure)	C_LB	C_gcg	C_UB	State
0,1,2	0.125000	2.079441	—	0.000000	0.000720	0.000000	STRONGLY MIXED
0,1,3	0.125000	2.079441	—	0.000000	0.000141	0.000000	STRONGLY MIXED
0,1,4	0.179494	1.834056	—	0.000000	0.004351	0.000000	STRONGLY MIXED
0,2,3	0.498930	0.701770	—	0.000000	0.000220	0.000720	MIXED
0,2,4	0.268683	1.605386	—	0.000000	0.000676	0.246747	MIXED
0,3,4	0.251866	1.654117	—	0.000000	0.000021	0.178688	MIXED
1,2,3	0.497966	0.709751	—	0.000000	0.000096	0.000000	MIXED
1,2,4	0.166176	1.915503	—	0.000000	0.000001	0.000000	STRONGLY MIXED
1,3,4	0.201332	1.800221	—	0.000000	0.000000	0.000000	MIXED
2,3,4	0.249009	1.448146	—	0.000000	0.003421	0.396798	MIXED

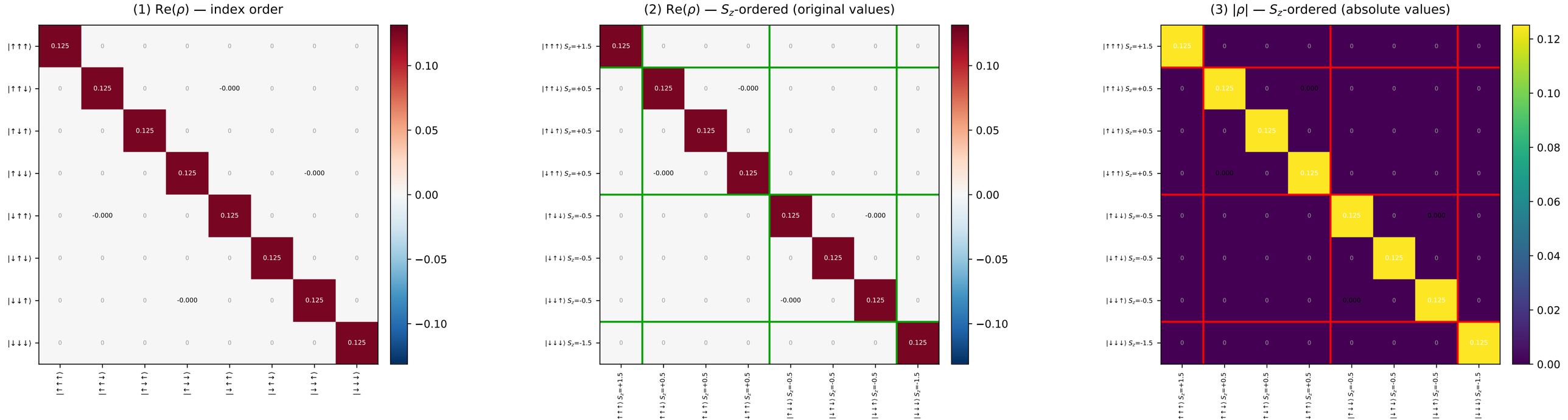
3-Body RDM (Sites 0,1,2) — same matrix in all three views



Trace=1.000000 Purity=0.125000 Entropy=2.079441 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

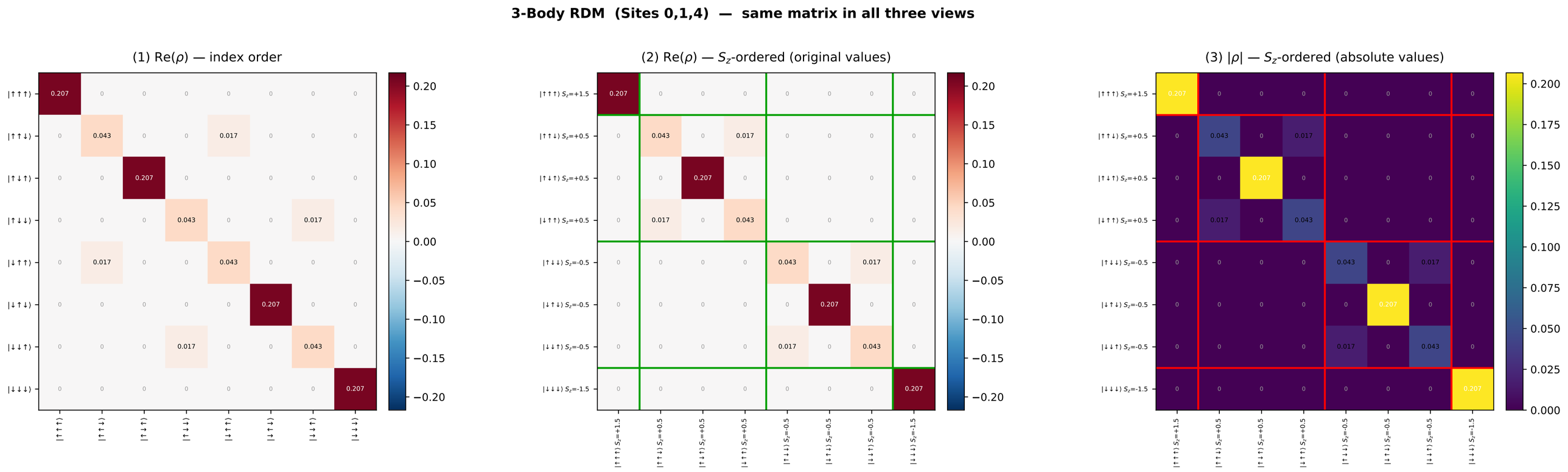
GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.000720 ≤ C_{UB} = 0.000000 verdict: GME

3-Body RDM (Sites 0,1,3) — same matrix in all three views

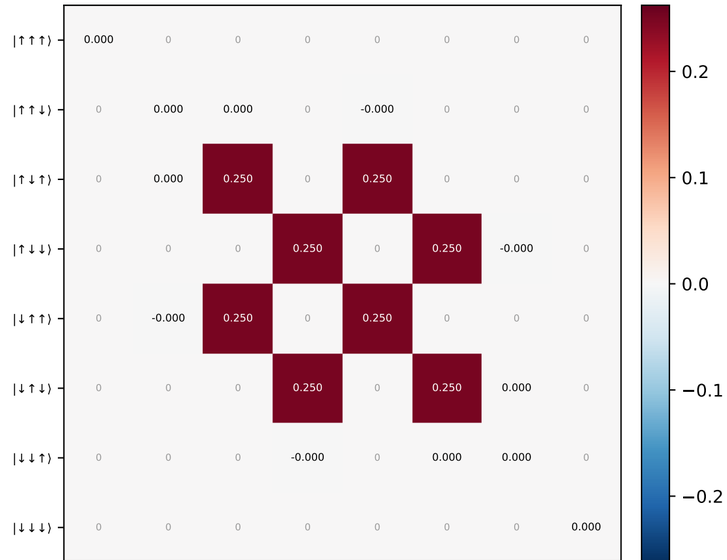


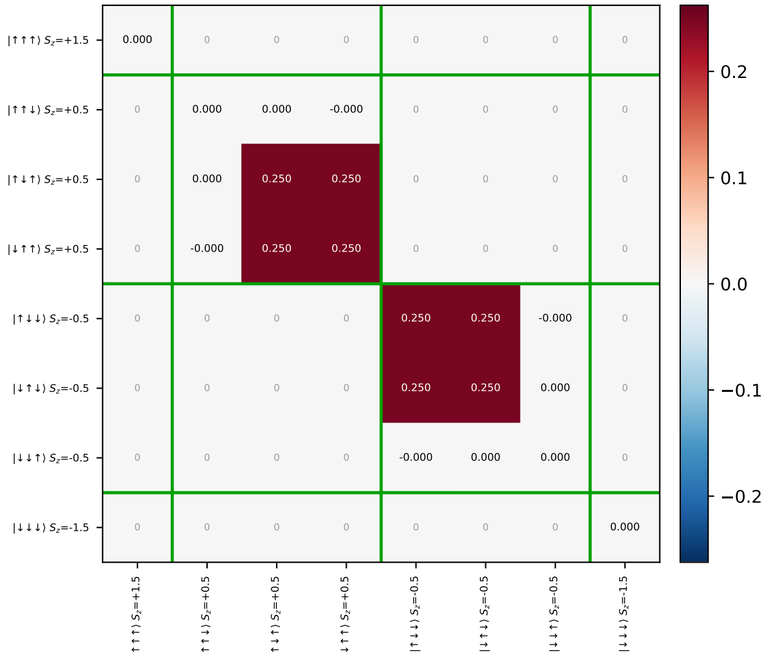
Trace=1.000000 Purity=0.125000 Entropy=2.079441 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

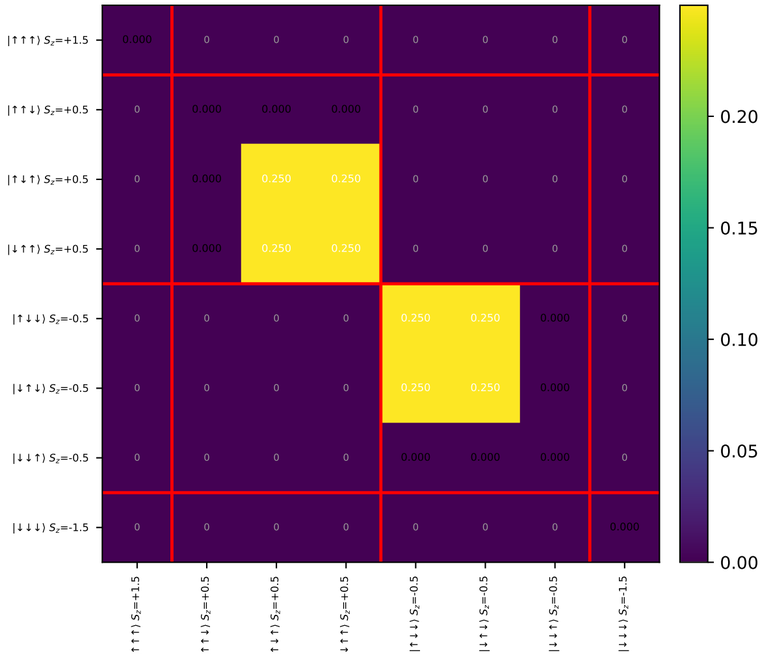
GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.000141 ≤ C_{UB} = 0.000000 verdict: GME


3-Body RDM (Sites 0,1,4) — same matrix in all three views
(1) Re(ρ) — index order
(2) Re(ρ) — S_z-ordered (original values)
(3) |ρ| — S_z-ordered (absolute values)
Trace=1.000000 Purity=0.179494 Entropy=1.834056 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)
GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.004351 ≤ C_{UB} = 0.000000 verdict: GME

3-Body RDM (Sites 0,2,3) — same matrix in all three views



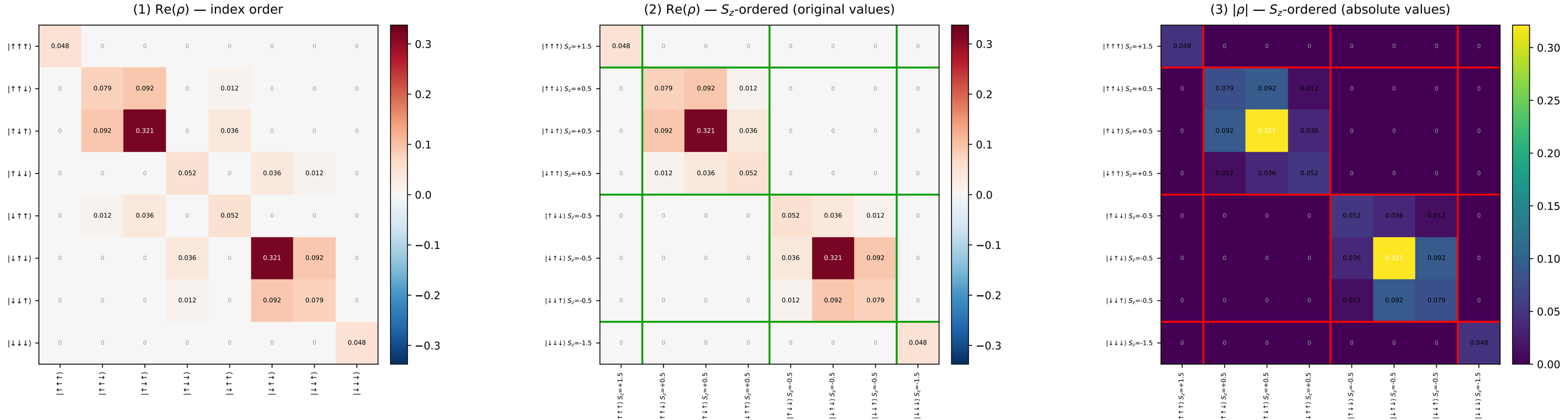




Trace=1.000000 Purity=0.498930 Entropy=0.701770 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.000220 ≤ C_{UB} = 0.000720 verdict: GME

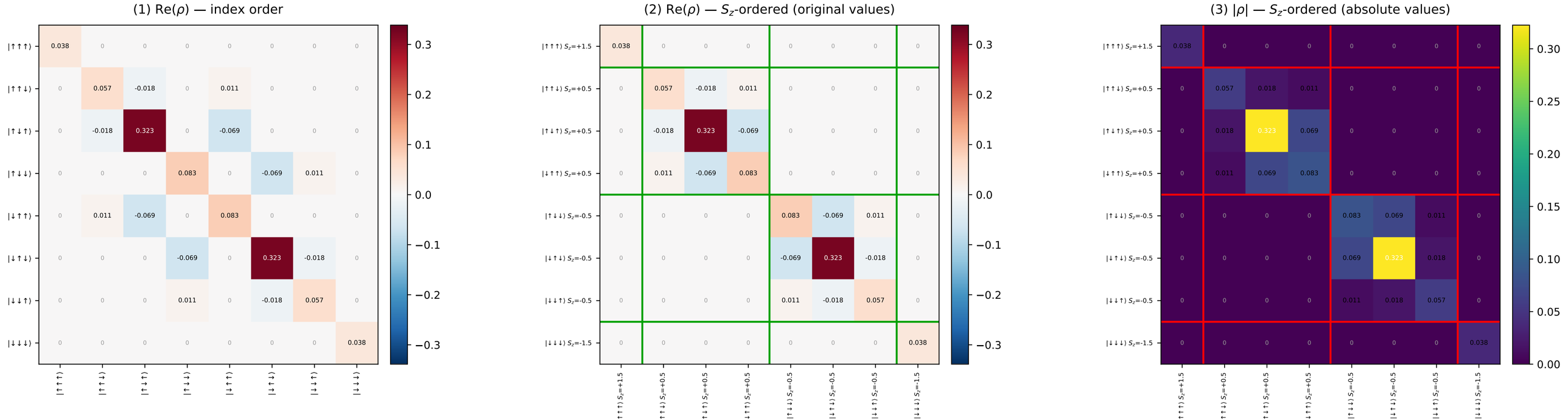
3-Body RDM (Sites 0,2,4) — same matrix in all three views



Trace=1.000000 Purity=0.268683 Entropy=1.605386 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 ($\Rightarrow S_z$ conserved)

GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.000676 ≤ C_{UB} = 0.246747 verdict: GME

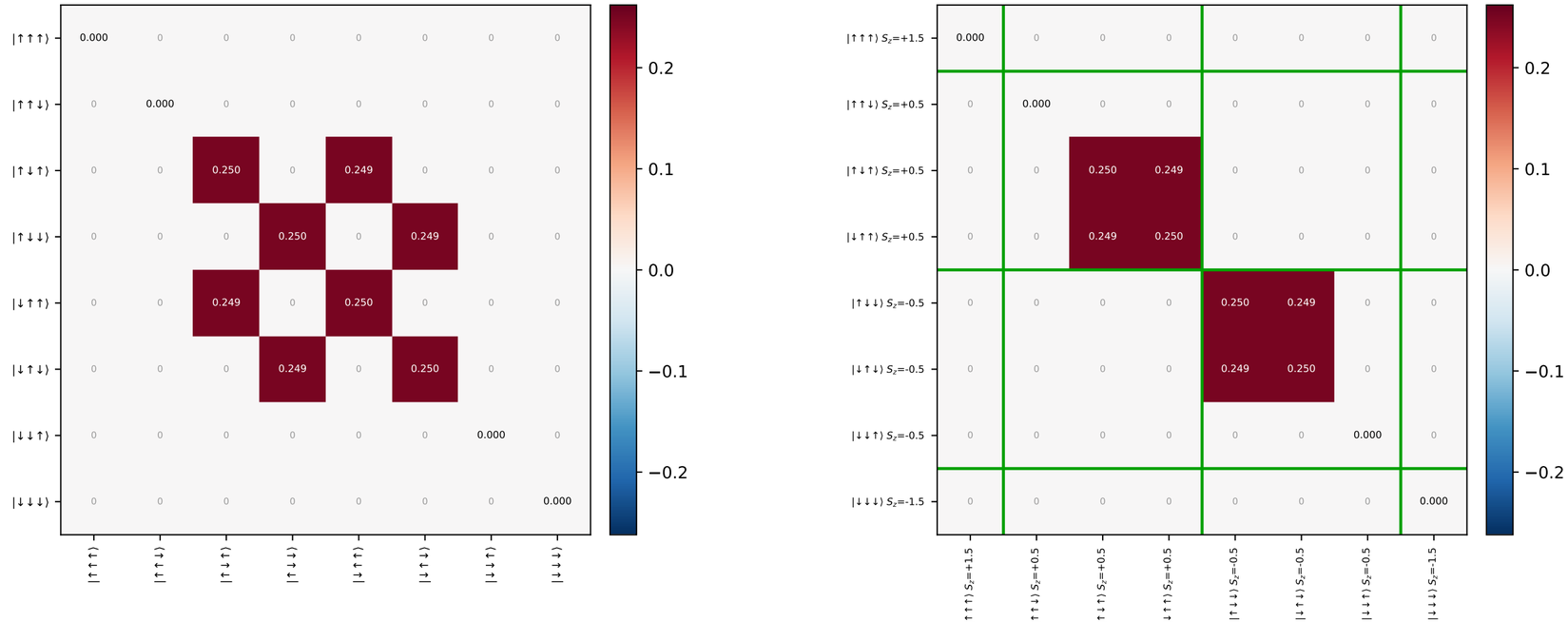
3-Body RDM (Sites 0,3,4) — same matrix in all three views



Trace=1.000000 Purity=0.251866 Entropy=1.654117 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

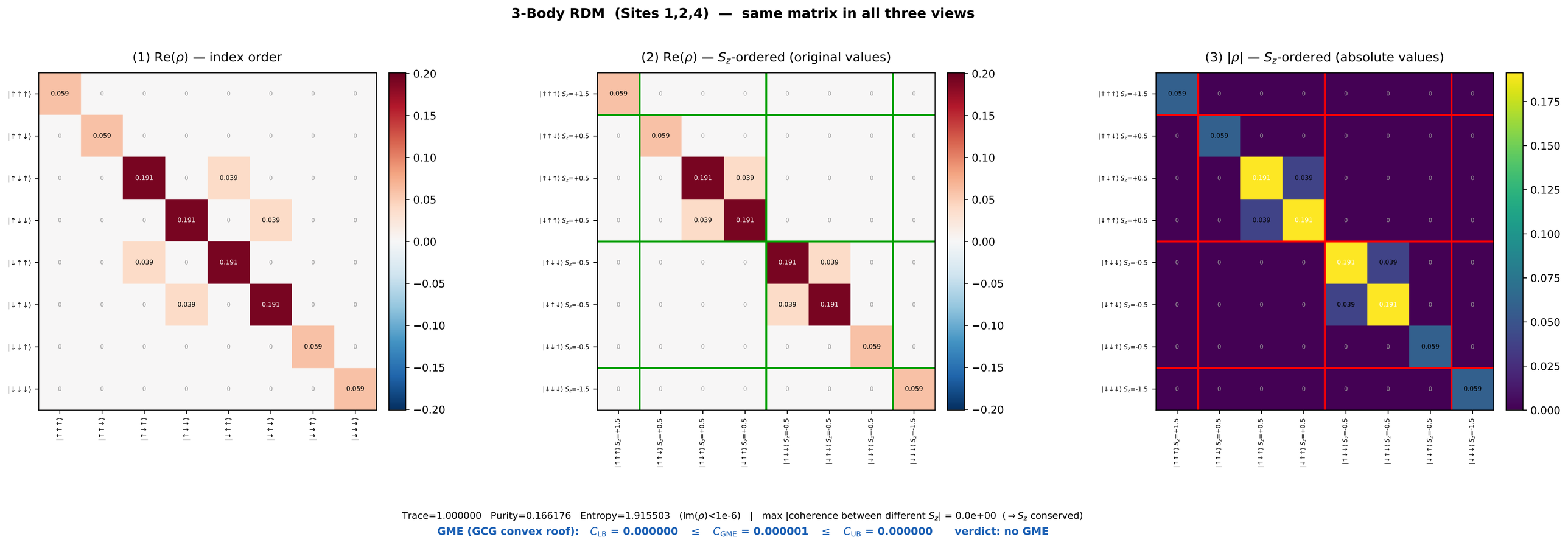
GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.000021 ≤ C_{UB} = 0.178688 verdict: GME

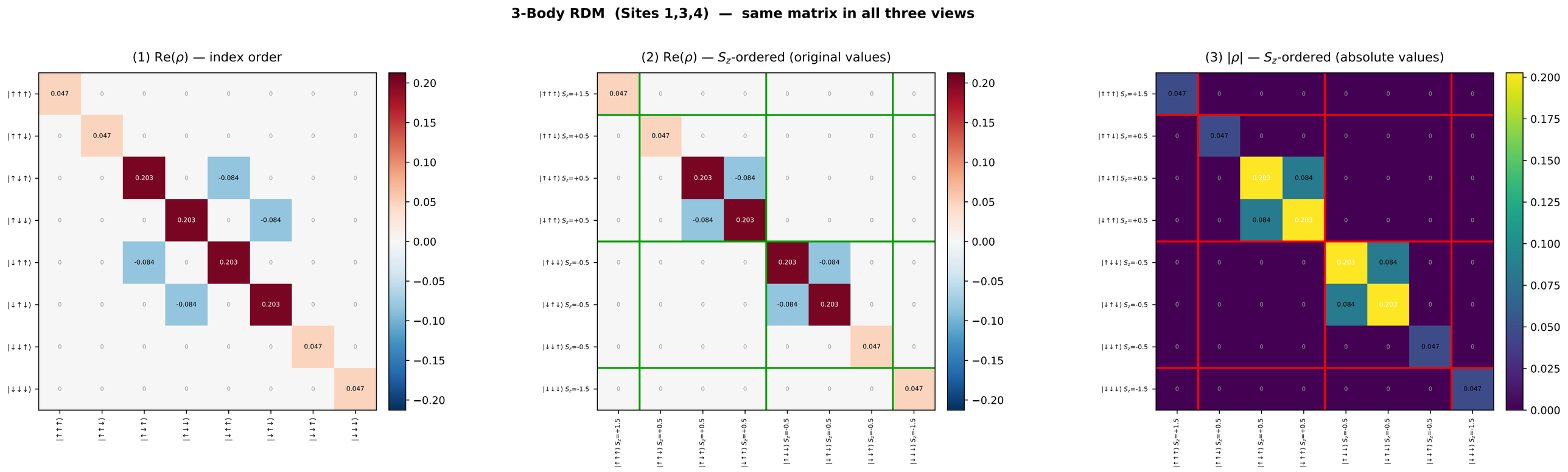
3-Body RDM (Sites 1,2,3) — same matrix in all three views



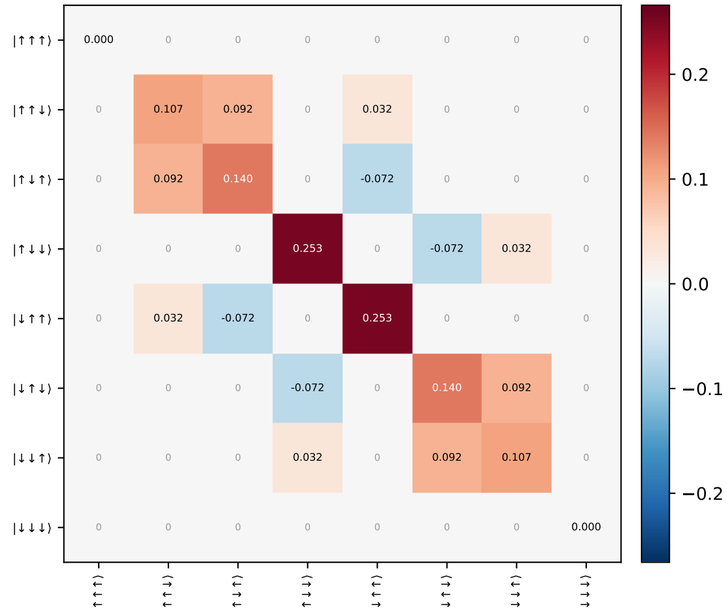
Trace=1.000000 Purity=0.497966 Entropy=0.709751 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

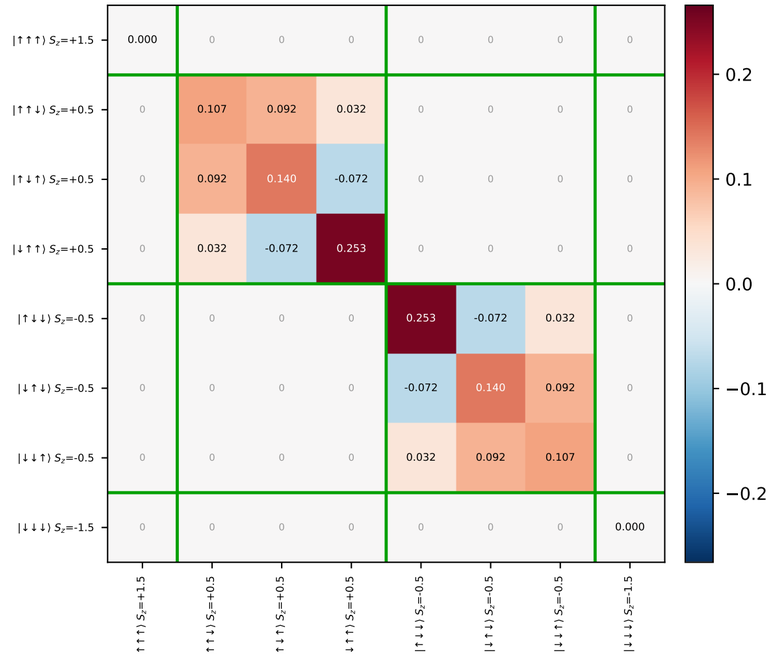
GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.000096 ≤ C_{UB} = 0.000000 verdict: GME

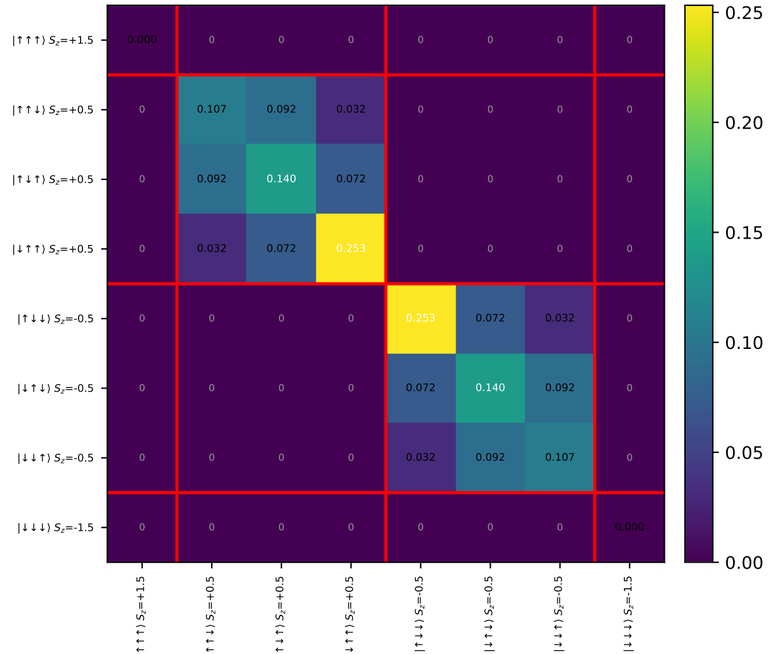

3-Body RDM (Sites 1,2,4) — same matrix in all three views
(1) Re(ρ) — index order
(2) Re(ρ) — S_z-ordered (original values)
(3) |ρ| — S_z-ordered (absolute values)
Trace=1.000000 Purity=0.166176 Entropy=1.915503 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)
GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.000001 ≤ C_{UB} = 0.000000 verdict: no GME


3-Body RDM (Sites 1,3,4) — same matrix in all three views
(1) Re(ρ) — index order
(2) Re(ρ) — S_z-ordered (original values)
(3) |ρ| — S_z-ordered (absolute values)
Trace=1.000000 Purity=0.201332 Entropy=1.800221 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)
GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.000000 ≤ C_{UB} = 0.000000 verdict: no GME

3-Body RDM (Sites 2,3,4) — same matrix in all three views







Trace=1.000000 Purity=0.249009 Entropy=1.448146 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.003421 ≤ C_{UB} = 0.396798 verdict: GME